

Method	LLM Size	Charades-STA			QVHighlight	
		$R@1_{(IOU=0.3)}$	$R@1_{(IOU=0.5)}$	$R@1_{(IOU=0.7)}$	mAP	$HIT@1$
MovieChat (Song et al., 2024a)	7B	8.8	2.9	1.3	11.7	16.1
GroundingGPT (Li et al., 2024c)	7B	-	29.6	11.9	-	-
VTimeLLM (Huang et al., 2024a)	7B	51.0	27.5	11.4	-	-
HawkEye (Wang et al., 2024f)	7B	50.6	31.4	14.5	-	-
TimeChat (Ren et al., 2024)	7B	-	32.2	13.4	14.5	23.9
ChatVTG (Qu et al., 2024)	7B	52.7	33.0	15.9	-	-
VideoChat2 (Li et al., 2024b)	7B	9.6	3.4	1.4	13.4	18.6
VideoChat-T	7B	69.9 (+60.3)	48.7 (+45.3)	24.0 (+22.6)	26.5 (+13.1)	54.1 (+35.5)
QD-DETR* (FT) (Moon et al., 2023b)	-	-	57.3	32.6	38.9	64.2
UnLoc-L* (FT) (Yan et al., 2023)	-	-	60.8	38.4	-	-
HawkEye (FT) (Wang et al., 2024f)	7B	72.5	58.3	28.8	-	-
Timechat (FT) (Ren et al., 2024)	7B	-	46.7	23.7	21.7	37.9
VideoChat-T (FT)	7B	79.4	67.1	43.0	27.0	55.3

Table 1: **Performance of VideoChat-T on temporal grounding and highlight detection tasks.** (FT) indicates the model fine-tuned on training set of the evaluation benchmark, with the respective text marked in gray. Classic supervised expert models are marked with *.

4 EXPERIMENTS

4.1 IMPLEMENTATION DETAILS

Built upon VideoChat2, we use UMT-L (Li et al., 2023c) and Mistral-7B (Jiang et al., 2023) as the video encoder and LLM, respectively. Except for the TAPE, all components are initialized from the pre-trained model of VideoChat2-Mistral. For the TAPE, we use random initialization, set the initial values of the final linear layer to zero, and freeze it during the first epoch of training. We set the frame count T for each clip to 8, so the number of clips K for a long video is equal to the total frame count divided by T . We fine-tune the model for 3 epochs using the TimePro with 349K instances and a general QA task dataset with 82K instances. To ensure the stability of model training, we use 192-frame input for the first epoch. In the second and third epochs, we unfreeze the TAPE and adjust the model input to 128 frames. All experiments are conducted on 16 A100 GPUs.

4.2 PERFORMANCE ON TEMPORAL GROUNDING

We evaluate our method using two commonly used temporal localization tasks, i.e., Temporal Grounding and Highlight Detection. The performance comparison between VideoChat-T and other models is shown in Table 1. Our method’s zero-shot performance surpasses all previous LLM-based methods and after fine-tuning, VideoChat-T even exceeds some classic expert models on the temporal grounding task.

Temporal Grounding. This task aims to identify the start and end timestamps of the video content described by the query sentence, using Charades-STA as the evaluation benchmark. VideoChat-T achieves an accuracy of 48.7 in the $R@1$ (IOU=0.5) metric, significantly surpassing the previous state-of-the-art MLLM method, namely TimeChat, by 16.5 points. Additionally, it outperforms the fine-tuned version of TimeChat on the training set of the evaluation benchmark by 2.0%. Furthermore, the performance of VideoChat-T fine-tuned on the evaluation benchmark training set reaches 67.1 $R@1$ at IoU=0.5, surpassing most state-of-the-art classic supervised expert models.

Highlight Detection. We use QVHighlights as the evaluation benchmark. For a given query, this task requires outputting all timestamps of highlight moments and their corresponding saliency scores. Since there could be many sparse highlight moments in a video, this task requires finer-grained video understanding at the frame level. VideoChat-T achieves mAP of 26.5, significantly surpassing the previous MLLM method of TimeChat by 13.0 points, and also outperforms its fine-tuned version by 4.8 points. We observe that after fine-tuning on the corresponding training set, VideoChat-T shows almost no performance improvement. This may be due to the bottleneck in language representation of LLMs. The Highlight Detection task requires outputting a (timestamp, saliency score) pair for each highlight moment, and a video may contain dozens of discrete highlight moments, making it challenging for the model to correctly respond with dozens to hundreds of numbers in a language format. The precise numerical saliency score output is very difficult for LLMs, and VideoChat-T can only respond well to queries with fewer highlight moments. Due to the

Method	LLM Size	Long Video				Short Video
		Egoschema		VideoMME		MVBench
		Subset	Full	w/o subs	w/o subs (Long)	Avg
VideoAgent (Wang et al., 2024c)	GPT-4	60.2	54.1	-	-	-
VideoAgent (Fan et al., 2024)	GPT-4	62.8	-	-	-	-
TimeChat (Ren et al., 2024)	7B	-	33.0	30.2	26.1	38.5
LLAMA-Vid (Li et al., 2023d)	7B	-	38.5	-	-	41.9
MovieChat (Song et al., 2024a)	7B	-	53.5	38.2	33.4	55.1
MovieChat+ (Song et al., 2024b)	7B	-	56.4	-	-	-
Chat-UniVi (Jin et al., 2024)	7B	-	-	40.6	35.8	-
VideoChat2 (Li et al., 2024b)	7B	63.6	54.4	39.5	33.2	60.4
VideoChat-T	7B	68.4 (+4.8)	60.0 (+5.6)	46.3 (+6.8)	41.9 (+8.7)	59.9 (-0.5)

Table 2: **Performance of VideoChat-T and other methods on video question answering tasks.** By upgrading VideoChat2 with TimeSuite, VideoChat-T demonstrates significant improvements across multiple long video benchmarks.

specific architectural design, classic supervised expert models have a natural advantage in handling such tasks, and VideoChat-T still has a performance gap compared to expert models.

4.3 PERFORMANCE ON GENERAL VIDEO QA

In addition to test the grounding ability of our VideoChat-T, we also want to verify its general video question answering performance. According to mainstream evaluation standards, we use both long video and short video QA to assess the general video understanding capability of VideoChat-T. Table 2 shows the performance of VideoChat-T on the video QA evaluation benchmarks.

Long Video QA. We use Egoschema (Mangalam et al., 2023) and VideoMME (Fu et al., 2024) to evaluate the long video capabilities of VideoChat-T. In conjunction with our proposed architectural improvements, we incremental fine-tune VideoChat2 using only 432K data points. VideoChat-T demonstrates outstanding performance on the Egoschema, achieving an accuracy of 68.4% on the test subset and 60.0% on the entire test set. Compared to VideoChat2, VideoChat-T obtains improvements of 4.8% and 5.6% on the subset and the full test set, respectively. Additionally, for the VideoMME benchmark, VideoChat-T achieves an accuracy of 46.3% by solely analyzing the visual content without using subtitles, representing a 6.8% improvement over VideoChat2. On the long video data division of VideoMME, VideoChat-T achieves an accuracy of 41.9%, which is an 8.7% improvement compared to VideoChat2. The upgraded VideoChat-T demonstrated significant performance improvements on long video QA benchmarks. This indicates the potential of leveraging grounding-centric video tasks to enhance the temporal awareness of MLLMs, thereby further improving long video understanding capabilities.

Short Video QA. We use MVBench (Li et al., 2024b) to evaluate the general short video understanding capabilities of VideoChat-T. VideoChat-T achieves an overall average accuracy of 59.9% on MVBench, which is a 0.5% decrease compared to VideoChat2. It is important to note that achieving minimal performance loss is a challenging task. According to previous experiences in the field of incremental learning (Van de Ven et al., 2022), models inevitably forget old knowledge while learning new knowledge. VideoChat2 is fine-tuned with 2M data, whereas VideoChat-T is fine-tuned with only 432K data, where 349K annotations are temporal grounding centric, resulting in only a 0.5% accuracy loss. Previous temporal MLLMs like TimeChat (Ren et al., 2024), although achieving strong temporal localization capabilities, yield much weaker general video QA capability, with an accuracy of only 38.5% on MVBench. This demonstrates that the design of our TimeSuite enhances new capabilities for the model while still preserving the original general video understanding capabilities. For a detailed analysis of the performance degradation of MVBench, please refer to Appendix F.2.

4.4 QUALITATIVE ANALYSIS

Figure 4 presents a qualitative comparison between our model and other methods. In the example on the left, VideoChat-T is capable of answering more complex long video reasoning questions. Our model accurately identifies the temporal location of the “light a cigarette” event and determines the correct key clue “the person in a white coat” based on the video content. This leads to the inference

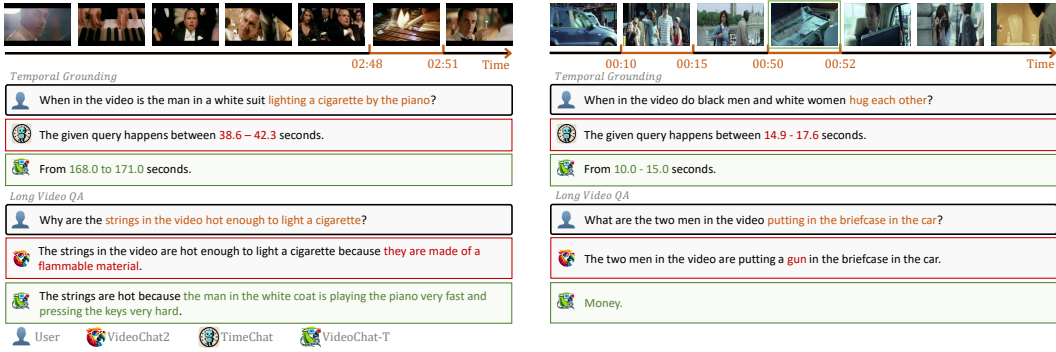


Figure 4: Qualitative comparison between VideoChat-T and other methods. VideoChat-T not only possesses temporal fine-grained perception capabilities but also can perform accurate long video reasoning. Green text indicates correct answers, while red text indicates inappropriate answers.

Model	Egoschema Full	VideoMME w/o subs	Charades-STA R@1 IOU=0.5	QVHighlights Hit@1
VideoChat-T (Ours)	60.0	46.3	48.7	54.1
w/o TAPE	59.1	45.9	47.1	50.4
w/o frz	59.0	45.2	52.4	53.7

Table 3: Performance results of the ablation study on the TAPE. Here, w/o adapter refers to removing our proposed TAPE, and w/o frz refers to not using the training method where the TAPE is frozen during the first epoch.

Model	Egoschema Full	VideoMME w/o subs	Charades-STA R@1 IOU=0.5	QVHighlights Hit@1
VideoChat-T(Ours)	60.0	46.3	48.7	54.1
r/w pooling	59.8	44.8	40.3	47.3
r/w clustering	59.5	45.0	39.8	40.1
w/o init	57.4	43.4	42.0	53.9

Table 4: Performance results of the ablation study on the Token Shuffle. Here, r/w refers to replacing Token Shuffle with the other component, and w/o init refers to removing the efficient initialization.

that “playing the piano very fast and pressing the keys very hard” are the true reasons. The example on the right demonstrates our model’s fine-grained perception ability. The appearance of “money in the briefcase” is very brief, and most models easily overlook this detail. Thanks to its strong fine-grained perception ability, our model precisely captures this visual content.

4.5 ABLATION STUDY

Role of TAPE. To verify the performance improvement brought by TAPE, ablation experiments were conducted. Table 3 lists the performance results of the conducted adapter-related ablation experiments. It can be observed that when the TAPE is removed, the model’s performance on long video understanding and temporal grounding benchmarks decreases. TAPE can adaptively embed positional encodings into video tokens, and the absence of TAPE leads to a certain loss in temporal awareness capability. When we unfroze the TAPE in the first epoch, the performance improved on the temporal grounding task but declined on the long video QA task. This is because the TAPE is highly suited for tasks with strong temporal dependencies. If unfrozen too early, the model may become biased towards fitting temporal grounding tasks. Freezing the TAPE during the first epoch allows the model to first optimize and learn a relatively generalized feature representation, thereby balancing the performance across different tasks.

Effectiveness of Token Shuffle. To verify the effectiveness of token shuffle, we conducted ablation experiments. Table 4 presents the results of these ablation experiments. We compared token shuffle with conventional methods such as pooling and clustering, and also observed the results after removing efficient initialization. When we replaced token shuffle with pooling or clustering methods, the model’s performance declined. This is because the efficient initialization of the linear layer in token shuffle makes the initial values of the module equivalent to average pooling, which gradually optimizes better solutions during training. Therefore, our method is inherently superior to pooling. On the other hand, clustering often fails to maintain the spatial/temporal consistency of the video, leading to temporal confusion. When we removed the efficient initialization of the linear layer, the negative impact of random initialization severely damaged the model’s original performance.

Effect of TimePro. We conducted ablation studies to evaluate the effectiveness of the TimePro data components. As shown in Table 5, by gradually adding subsets of TimePro, we observed the model’s performance changes across various temporal grounding-centric instruction-tuning data. As we pro-